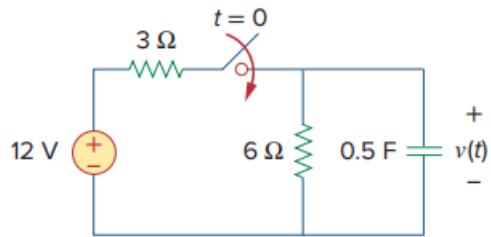


Problem

For the circuit in Fig. 7.66, use *Pspice* to find $v(t)$ for $t > 0$.

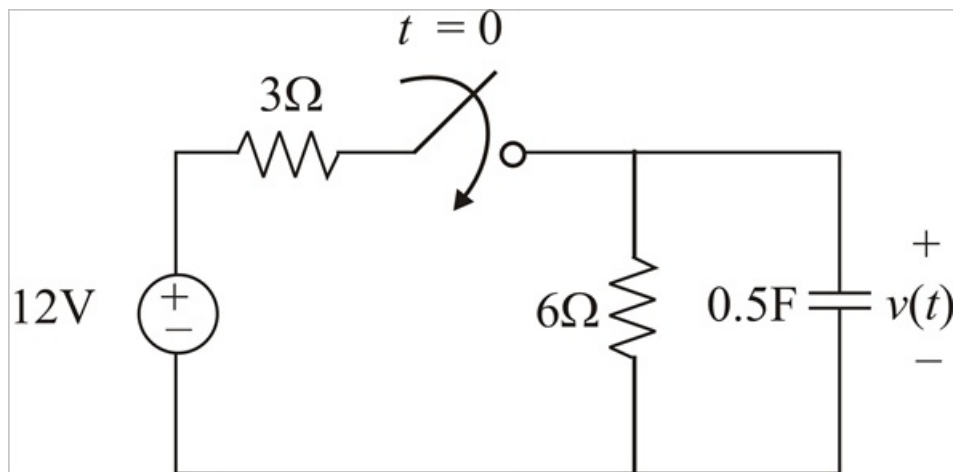
Fig. 7.66:



Step-by-step solution

Step 1 of 6

Given circuit diagram



Step 2 of 6

For $t < 0$ switch is in open condition, so capacitor initial voltage is

$$v(0) = 0V$$

For $t > 0$ switch is in closed condition, so capacitor final voltage is

$$v(\infty) = 12 \times \left(\frac{6}{3+6} \right)$$

$$v(\infty) = 8V$$

Equivalent resistance seen by the capacitor is

$$R_{eq} = 3\Omega \parallel 6\Omega$$

$$R_{eq} = \frac{3 \times 6}{3+6}$$

$$R_{eq} = 2\Omega$$

Time constant of the circuit is

$$\tau = R_{eq} C$$

$$\tau = (2)(0.05)$$

$$\tau = 1\text{sec}$$

Step 3 of 6

Therefore capacitor voltage is

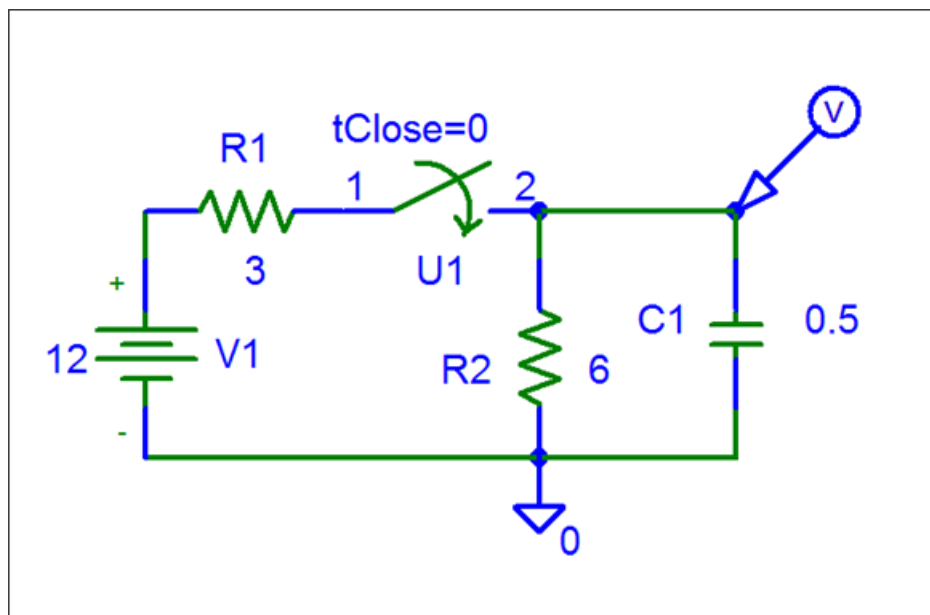
$$v(t) = v(\infty) + \left[v(0) - v(\infty) e^{-\frac{t}{\tau}} \right]$$

$$v(t) = 8 + \left[0 - 8e^{-\frac{t}{1}} \right]$$

$$\boxed{v(t) = 8(1 - e^{-t}) \text{ V}} \quad \text{for } t > 0$$

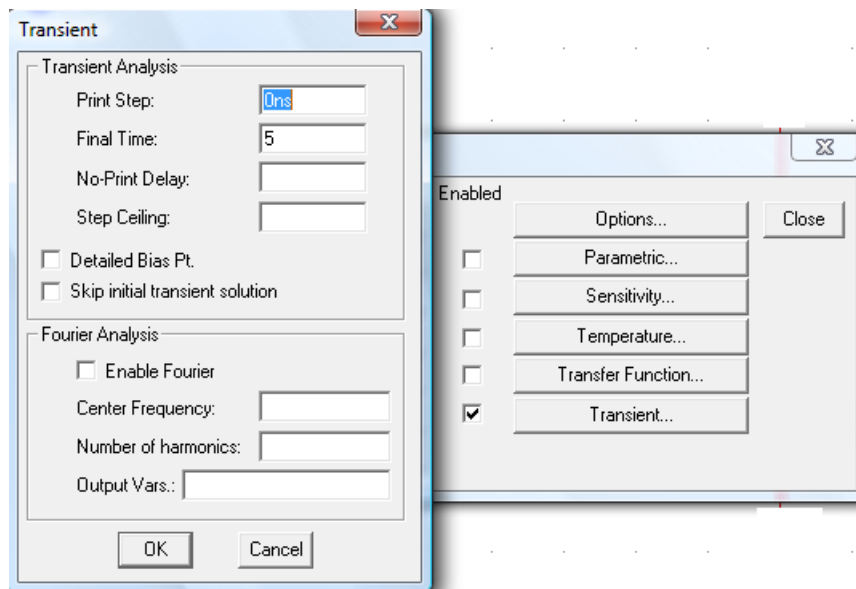
Step 4 of 6

Construct PSPICE circuit as shown in below figure

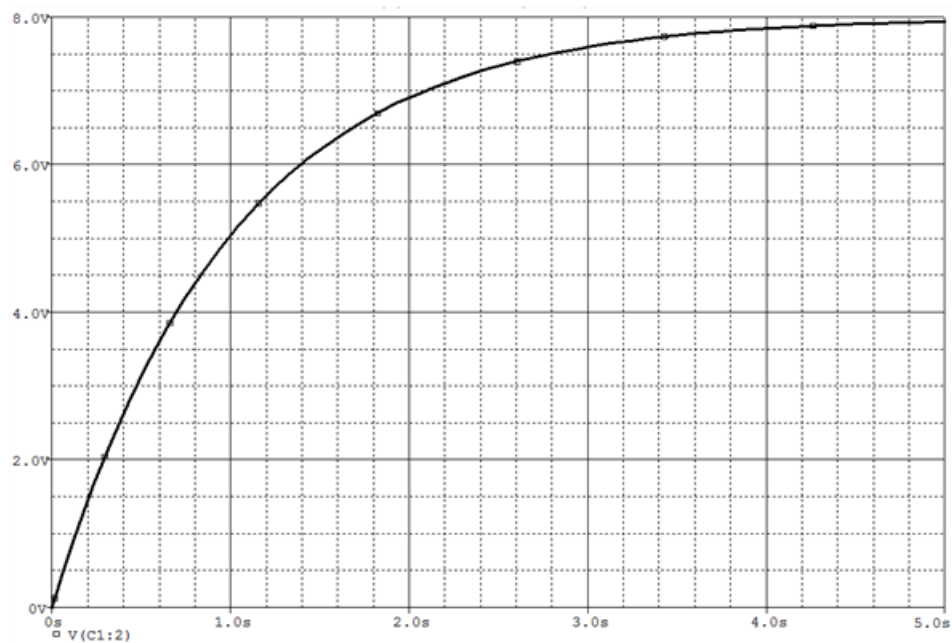


Step 5 of 6

And select **analysis/setup/transient** and enter parameters as shown in fig



Then press F-11 OR analysis/simulate then we will get the required PSPICE plot



Step 6 of 6

Checking:

Take $t = 1.25\text{sec}$

$$v(1.25\text{sec}) = 8(1 - e^{-1.25})$$

$$v(1.25\text{sec}) = 5.708\text{V}$$

From the above graph $v(1.25\text{sec}) = 5.708\text{V}$

Hence proved.